Mathematical Modelling and Simulation of Quantum Gate Operations Using Qiskit: Project Report

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Abstract

This report derives, simulates and compares the action of the fundamental single-qubit gates (Pauli-X, Pauli-Y, Pauli-Z, Hadamard and the rotations Rx, Ry, Rz) using Qiskit. Each analytical prediction, computed independently with NumPy, was compared with Qiskit statevector and shot-based simulations on four input states, a 73-angle rotation sweep, three gate-order comparisons and a finite-shot experiment. The gates were unitary to 3.2×10^{-16}, and simulated states matched the predictions with fidelity at least $1 - 9\times10^{-16}$. Applying H and Z, or X and H, in opposite orders gave orthogonal states, even though the computational-basis probabilities on |0⟩ were identical, and the identity HZH = X was confirmed. Finite-shot estimation error followed the binomial N^(−1/2) law, falling from 0.052 at 100 shots to 0.005 at 10,000.

Keywords: quantum computing; quantum gates; mathematical modelling; Qiskit; Bloch sphere; unitary transformations.

1. Introduction

Quantum gates are unitary operators that transform the state of a qubit, and their behavior is governed entirely by linear algebra, complex vector spaces and probability theory [1]. A qubit can exist in a coherent superposition of the computational basis states |0⟩ and |1⟩, and a circuit is a sequence of gates acting on such states. The Qiskit framework provides tools for constructing circuits, simulating state evolution and visualizing results [2].

1.1 Problem statement

Quantum gates are often introduced through circuit diagrams and code, with little emphasis on the matrix operations underneath. A learner can build a circuit without understanding how a gate matrix transforms a state vector. This project connects the two: each gate is derived analytically, simulated numerically, and the results are compared.

1.2 Aim

To derive and numerically verify the matrix action of fundamental single-qubit gates, visualize state evolution on the Bloch sphere, investigate angle-dependent measurement probabilities, test gate-order dependence, and compare ideal probabilities with finite-shot sampling.

1.3 Objectives

1. Derive the matrix action of the I, X, Y, Z and H gates and verify that each is unitary.
2. Compare analytically calculated gate outputs with Qiskit state vectors on four input states.
3. Visualize state evolution on the Bloch sphere.
4. Examine how Rx(θ), Ry(θ) and Rz(θ) change Bloch coordinates and measurement probabilities across a sweep of angles.
5. Test the order dependence of selected gate sequences.
6. Compare ideal probabilities with finite-shot estimates.

1.4 Research questions

1. Do Qiskit state vectors reproduce analytically calculated gate transformations?
2. How do Rx(θ), Ry(θ) and Rz(θ) change Bloch coordinates and measurement probabilities?
3. Which gate sequences are order dependent, and can computational-basis measurements distinguish their outputs?
4. How do finite-shot probability estimates converge toward ideal probabilities?

2. Mathematical framework

Each gate is a 2×2 unitary matrix; its action on |0⟩ and |1⟩ is read from its first and second columns.

2.1 Qubit and gate transformation

A general pure state is a unit vector in $\mathbb{C}^2$:

```
|\psi\rangle = \alpha|0\rangle + \beta|1\rangle =
\begin{pmatrix}\alpha\\\beta\end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1
```

A gate U acts as |ψ_out⟩ = U|ψ_in⟩ and must satisfy the unitarity condition:

```
U^{\dagger}U = I
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2.2 Pauli and Hadamard gates

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X = \begin{pmatrix}0&1\\1&0\end{pmatrix},\quad Y = \begin{pmatrix}0&-
i\\i&0\end{pmatrix},\quad Z = \begin{pmatrix}1&0\\0&-1\end{pmatrix},\quad H =
\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}
```

Gate	On |0⟩	On |1⟩	Effect
X	|1⟩	|0⟩	Bit flip
Y	i|1⟩	−i|0⟩	Bit flip with phase
Z	|0⟩	−|1⟩	Phase flip
H	(|0⟩ + |1⟩)/√2	(|0⟩ − |1⟩)/√2	Equal superposition

Unitarity is verified by direct multiplication. X, Y, Z and H are all Hermitian ($U^\dagger = U$), so the check reduces to $U^2 = I$. For Y, the entries of $Y^\dagger Y$ are $(-i)(i) = 1$ on the diagonal and 0 off it. For H, $(1/2)[[1+1, 1-1],[1-1, 1+1]] = I$.

2.3 Rotation gates

The rotations are exponentials of the Pauli matrices. Because $\sigma^2 = I$, the power series splits into even terms (giving $\cos(\theta/2)\cdot I$) and odd terms (giving $-i\cdot\sin(\theta/2)\cdot\sigma$):

```
R_{\sigma}(\theta) = e^{-i\theta\sigma/2} = \cos(\theta/2)\,I -
i\sin(\theta/2)\,\sigma
```

```
R_x(\theta)=\begin{pmatrix}\cos\frac{\theta}{2}&-i\sin\frac{\theta}{2}\\-
i\sin\frac{\theta}{2}&\cos\frac{\theta}{2}\end{pmatrix},\quad
R_y(\theta)=\begin{pmatrix}\cos\frac{\theta}{2}&-\sin\frac{\theta}
{2}\\\sin\frac{\theta}{2}&\cos\frac{\theta}{2}\end{pmatrix},\quad
R_z(\theta)=\begin{pmatrix}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{pmatrix}
```

Each is unitary because it is built from $\cos^2(\theta/2) + \sin^2(\theta/2) = 1$. At $\theta = \pi$ each reduces to a Pauli gate up to a global phase: $R_x(\pi) = -iX$, $R_y(\pi) = -iY$, $R_z(\pi) = -iZ$.

2.4 Measurement probabilities

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$. Applying $R_y(\theta)$ to $|0\rangle$ gives the reference solution used throughout this report:

$$R_y(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle, \qquad P(0)=\cos^2\frac{\theta}{2}, \quad P(1)=\sin^2\frac{\theta}{2}$$

2.5 Bloch sphere

Up to a global phase, any pure state can be written with polar angle θ and azimuth φ, with Bloch vector r:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle, \qquad \vec{r} = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)$$

R_x, R_y and R_z rotate this vector by θ about the x, y and z axes respectively.

2.6 Gate sequences

For gates applied in order U_1 then U_2, the final state is $U_2U_1|\psi\rangle$. In general $U_1U_2 \neq U_2U_1$, so gate order matters.

3. Methodology

The study is a single Jupyter notebook that derives each prediction with independent NumPy matrix calculations and then compares it with a Qiskit simulation. All computation was done on classical simulators, and no quantum hardware results are claimed.

3.1 Software and reproducibility

The notebook uses Python with Qiskit (1.2 or later), Qiskit Aer (0.15 or later), NumPy, SciPy, pandas and Matplotlib. A fixed random seed of 2026 is used throughout, and every table and figure is saved to a results folder (CSV and PNG files) that is exported as a ZIP archive.

3.2 Experimental design

Five experiments were run, each comparing a NumPy prediction with a Qiskit result.

Experiment	Qiskit method	Setup
Unitarity	NumPy only	$U^\dagger U - I$ for I, X, Y, Z, H
Fundamental gates	State vector (no measurement)	X, Y, Z, H on $\|0\rangle$, $\|1\rangle$, $\|+\rangle$, $\|+i\rangle$
Rotation sweep	State vector	Rx, Ry, Rz at 73 angles from 0 to 2π, on $\|0\rangle$ and $\|+\rangle$
Gate order	State vector	H→Z vs Z→H, X→H vs H→X, and H→Z→H vs X, on $\|0\rangle$ and $\|+\rangle$
Finite shots	Aer shot-based simulator	H$\|0\rangle$ with 100, 500, 1000, 5000 and 10000 shots, 40 repeats each

Several input states are used because $|0\rangle$ alone hides phase changes: Z and Rz leave the probabilities of $|0\rangle$ unchanged, so superposition inputs such as $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ are needed to see them.

3.3 Verification metrics

Global phase does not change physical predictions, so states are compared by fidelity rather than component by component. The Bloch vector of a state $[\alpha, \beta]$ is computed from its amplitudes:

$$F = |\langle\psi_{\text{theory}}|\psi_{\text{Qiskit}}\rangle|^2, \qquad r_x = 2\,\mathrm{Re}(\alpha^*\beta),\quad r_y = 2\,\mathrm{Im}(\alpha^*\beta),\quad r_z = |\alpha|^2-|\beta|^2$$

The notebook also reports the L1 probability error (sum of absolute differences of the two outcome probabilities) and the L2 distance between Bloch vectors. For finite shots, $K \sim \text{Binomial}(N, p)$ and the estimate is $\hat{p} = K/N$, with predicted standard error:

$$\mathrm{SE} = \sqrt{p(1-p)/N}$$

The pre-specified expectations were unitarity residuals near machine precision, ideal state fidelity near 1, and shot errors consistent with binomial sampling.

4. Results: fundamental gates

Qiskit reproduced all 16 gate-and-input combinations: every fidelity was within 9×10^{-16} of 1, and the probability error and Bloch-vector error were 0 in every case.

4.1 Unitarity

The Frobenius norm $\|U^\dagger U - I\|$ was exactly 0 for I, X, Y and Z, and 3.16×10^{-16} for H. The Hadamard residual comes from the rounding of $1/\sqrt{2}$ and is at machine precision.

4.2 Gate action on four input states

Output states are given up to a global phase. P(0) and P(1) are the exact Qiskit statevector probabilities, which equal the theoretical values.

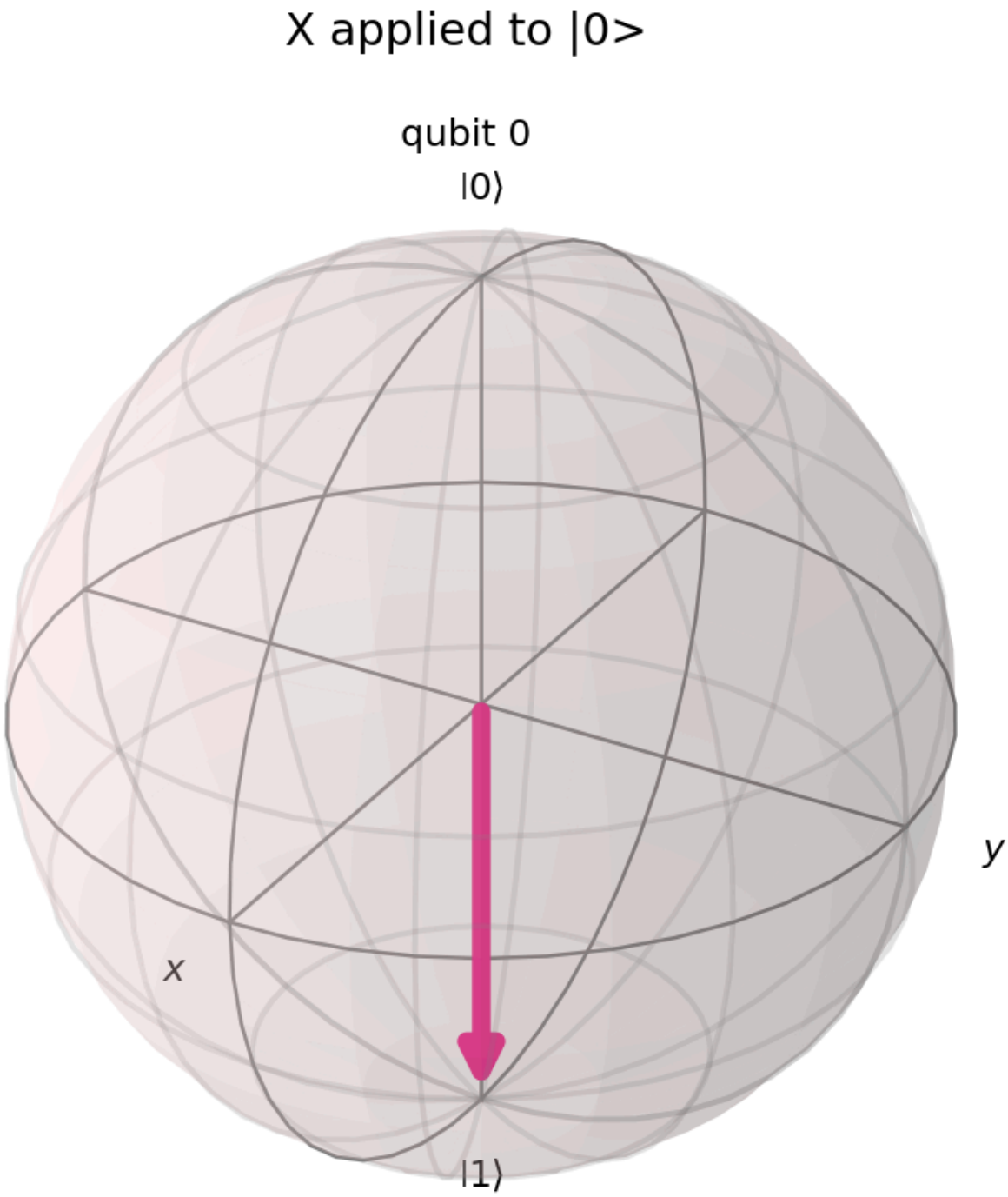
Input	Gate	Output state	P(0)	P(1)
$\|0\rangle$	X	$\|1\rangle$	0	1
$\|0\rangle$	Y	$\|1\rangle$	0	1
$\|0\rangle$	Z	$\|0\rangle$	1	0
$\|0\rangle$	H	$\|+\rangle$	0.5	0.5
$\|1\rangle$	X	$\|0\rangle$	1	0
$\|1\rangle$	Y	$\|0\rangle$	1	0
$\|1\rangle$	Z	$\|1\rangle$	0	1
$\|1\rangle$	H	$\|-\rangle$	0.5	0.5
$\|+\rangle$	X	$\|+\rangle$	0.5	0.5
$\|+\rangle$	Y	$\|-\rangle$	0.5	0.5
$\|+\rangle$	Z	$\|-\rangle$	0.5	0.5
$\|+\rangle$	H	$\|0\rangle$	1	0
$\|+i\rangle$	X	$\|-i\rangle$	0.5	0.5
$\|+i\rangle$	Y	$\|+i\rangle$	0.5	0.5
$\|+i\rangle$	Z	$\|-i\rangle$	0.5	0.5

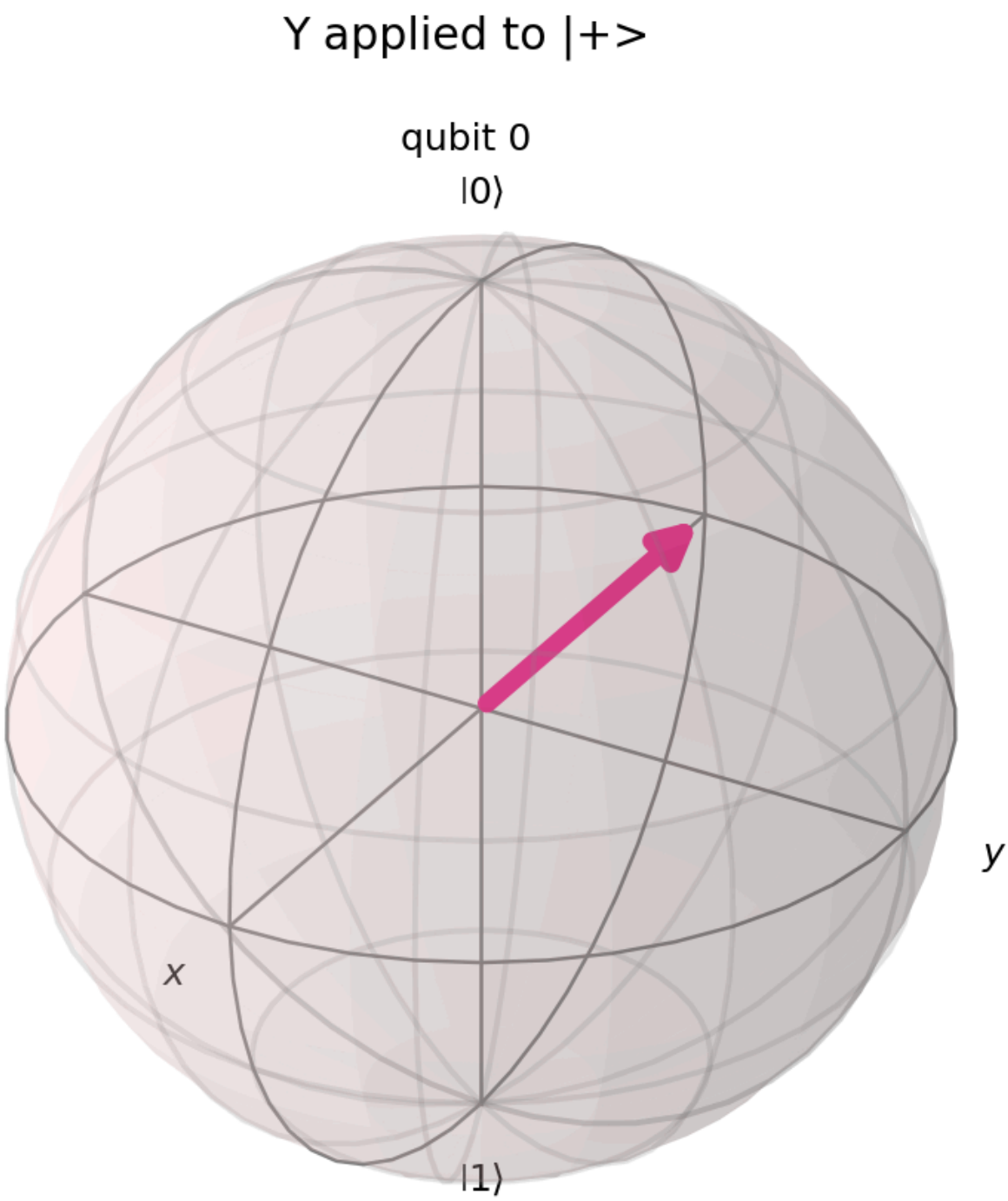
Input	Gate	Output state	P(0)	P(1)
$\|+i\rangle$	H	$\|-i\rangle$	0.5	0.5

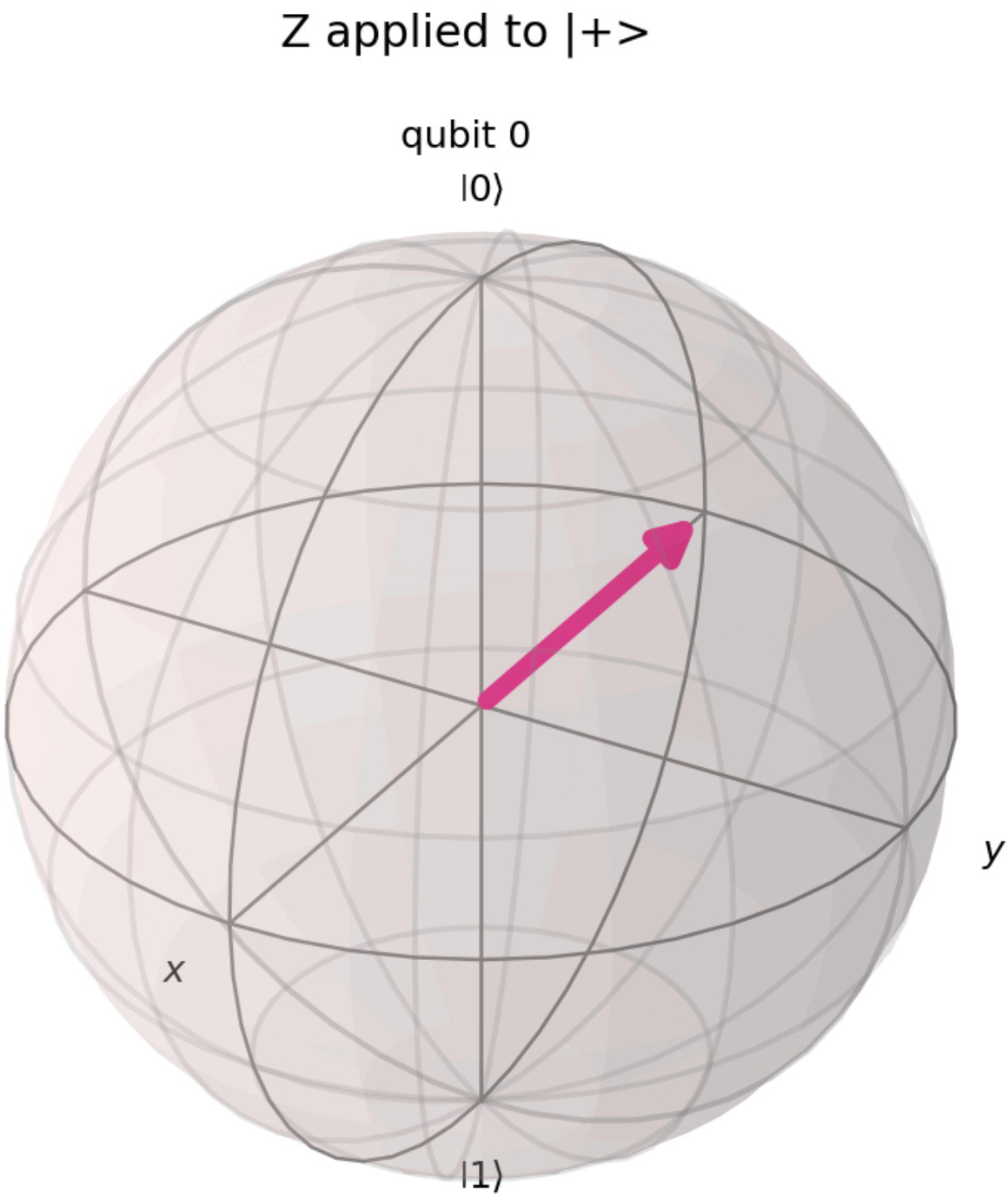
The table shows why several inputs were needed. Applied to $|0\rangle$ and $|1\rangle$, Z changes only the sign of $|1\rangle$, so the computational-basis probabilities look unchanged. Applied to $|+\rangle$, Z (and Y) turns $|+\rangle$ into $|-\rangle$: the Z-basis probabilities stay at 0.5, 0.5, yet the state has moved to the opposite side of the Bloch sphere. Applying H to $|+\rangle$ returns $|0\rangle$ with probability 1, which is interference between the two amplitudes.

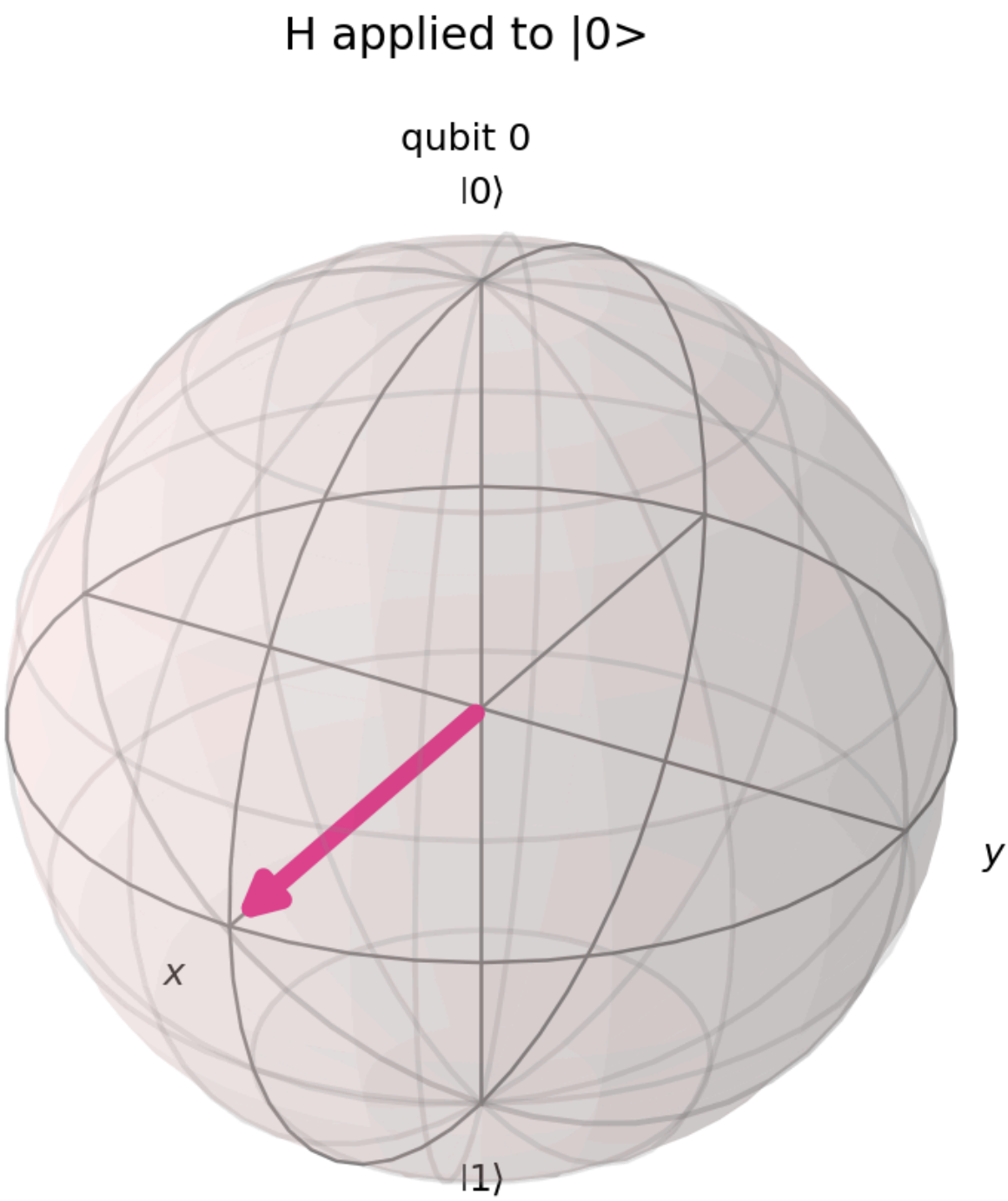
4.3 Bloch sphere visualization

Four representative transformations were plotted. The arrow is the Bloch vector of the output state.


X applied to |0>
qubit 0
|0⟩
y
x
|1⟩


Y applied to |+>
qubit 0
|0⟩
y
x
|1⟩


Z applied to |+>
qubit 0
|0⟩
y
x
|1⟩



X maps the north pole to the south pole, H moves the pole onto the equator at +x, and both Y and Z move $|+\rangle$ to $-x$. Y and Z give the same physical state here, since $Y|+\rangle = -i|-\rangle$ and $Z|+\rangle = |-\rangle$ differ only by a global phase.

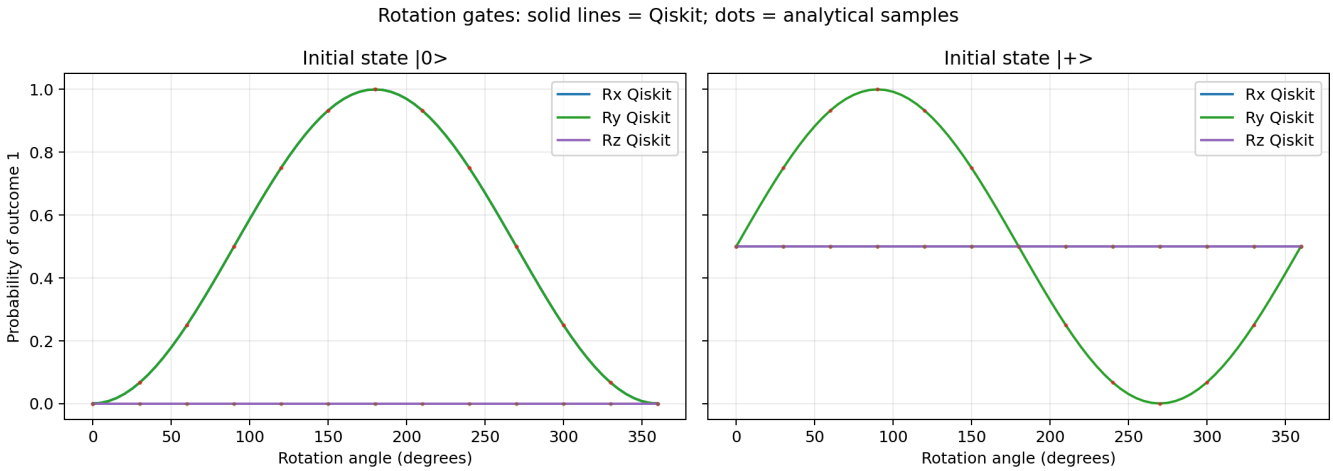
5. Results: rotation gates

Across all 438 sweep points (3 axes × 73 angles × 2 inputs), the Qiskit statevectors matched the closed-form rotations with minimum fidelity 1.0 and maximum probability error 0.

5.1 Measurement probabilities

The notebook sweeps the angle in 5° steps from 0° to 360°. The table shows P(1) at four angles, read from the saved sweep data.

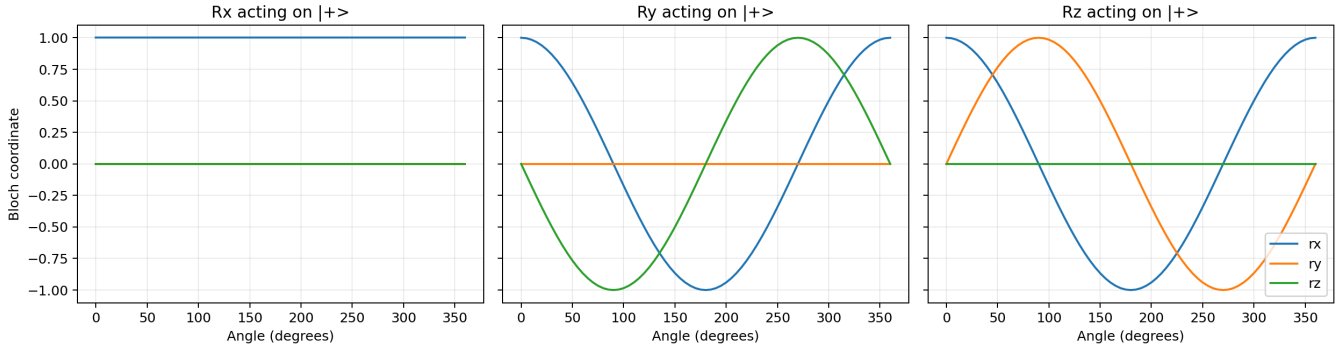
Input	Gate	0°	90°	180°	270°
|0⟩	Rx	0	0.5	1	0.5
|0⟩	Ry	0	0.5	1	0.5
|0⟩	Rz	0	0	0	0
|+⟩	Rx	0.5	0.5	0.5	0.5
|+⟩	Ry	0.5	1	0.5	0
|+⟩	Rz	0.5	0.5	0.5	0.5



On |0⟩, Rx and Ry give the same curve, $\sin^2(\theta/2)$, which is why the two lines overlap in the left panel. Rz is flat at 0 because |0⟩ is an eigenstate of Z. On |+⟩, Ry gives $P(1) = (1 + \sin\theta)/2$, while Rx and Rz are flat at 0.5 (the lines overlap in the right panel). The analytical dots lie on the Qiskit curves throughout.

5.2 Bloch coordinate evolution

Rz changes the state even where the probabilities do not, and the Bloch coordinates show this.



On |+⟩, which lies on the +x axis, the three rotations behave as follows.

- **Rx** leaves the vector at (1, 0, 0), because |+⟩ is an eigenstate of X.
- **Ry** rotates it in the x–z plane: rx = cos θ and rz = −sin θ. At 90° the vector is at (0, 0, −1), and at 270° it is at (0, 0, 1).
- **Rz** rotates it around the equator: rx = cos θ and ry = sin θ. At 90° it is at (0, 1, 0), and at 180° at (−1, 0, 0).

The same Rz that is invisible in the probabilities of |0⟩ therefore moves |+⟩ completely around the equator. At 360° every Bloch vector returns to its start, while the statevector itself returns only up to a global phase of −1.

6. Results: gate order and non-commutativity

Changing the order of H with Z, or of X with H, produces perfectly orthogonal states (fidelity 0), while the identity HZH = X holds to machine precision.

If A is applied first and B second, the output is BA|ψ⟩. The commutator is [A, B] = AB − BA, and for H and Z it has Frobenius norm ‖HZ − ZH‖ = 2, so the two gates do not commute.

Input	Sequence A	Sequence B	Fidelity of A vs B	Difference in P(1)
|0⟩	H → Z	Z → H	0	0
|+⟩	H → Z	Z → H	0	1
|0⟩	X → H	H → X	0	0
|+⟩	X → H	H → X	0	1

Input	Sequence A	Sequence B	Fidelity of A vs B	Difference in P(1)
$\|0\rangle$	H → Z → H	X	1 (to 4×10^{-16})	4×10^{-16}
$\|+\rangle$	H → Z → H	X	1 (to 7×10^{-16})	1×10^{-16}

In every row, the Qiskit statevector matched the NumPy prediction for both sequences with fidelity at least 0.9999999999999991.

Three conclusions follow from the table.

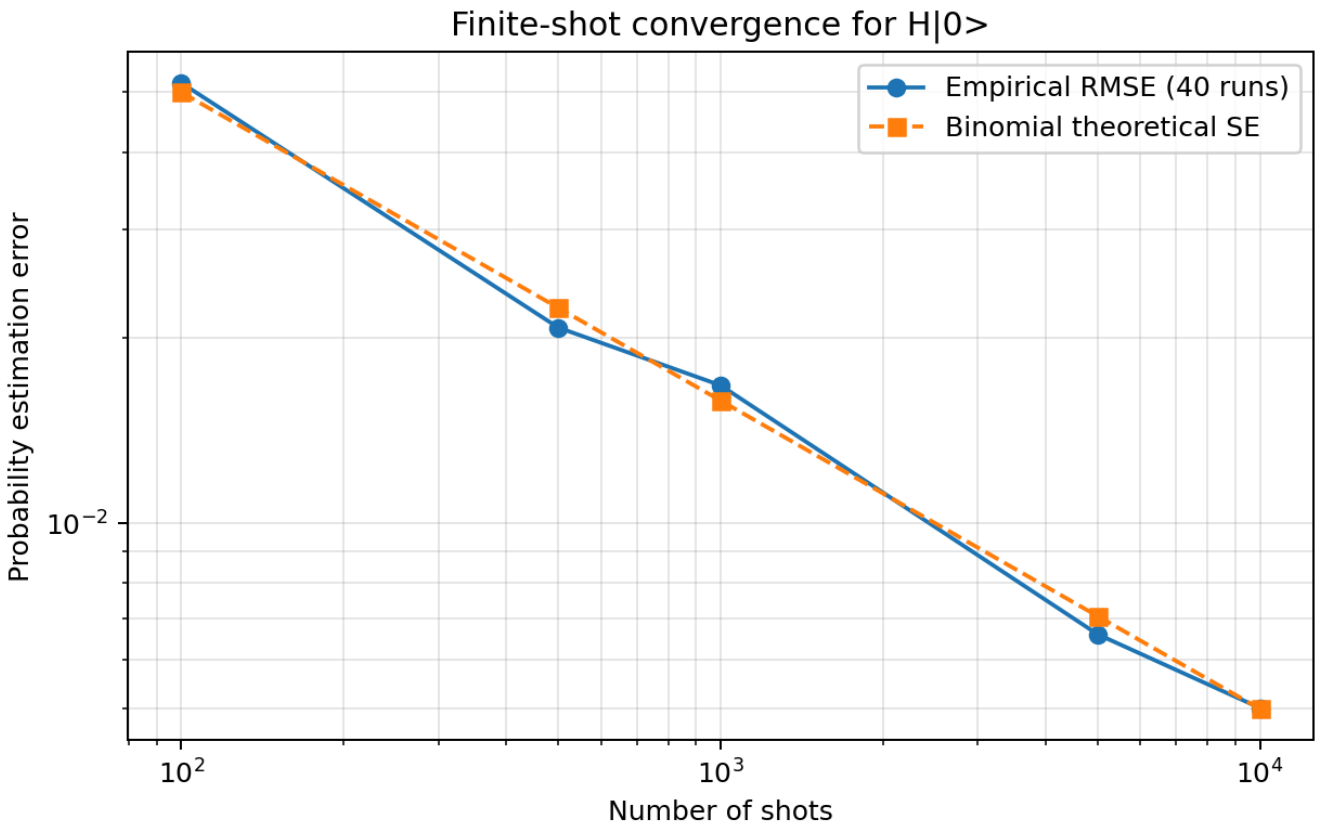
- **Order matters.** For H with Z and X with H, the two orders give orthogonal states on both inputs.
- **Z-basis probabilities can hide the difference.** On $|0\rangle$, A and B have identical probabilities (P(1) difference 0) yet fidelity 0: H → Z gives $|-\rangle$ and Z → H gives $|+\rangle$. A measurement in the X basis, obtained by applying H before measuring, would separate them. On $|+\rangle$ the probabilities also differ, by 1.
- **A circuit identity is confirmed.** Sandwiching Z between two Hadamards gives X, so H → Z → H and X act identically on both inputs.

7. Results: finite-shot measurement

The root-mean-square error of the estimated P(1) for H$|0\rangle$ fell with the number of shots in line with the binomial prediction, from 0.052 at 100 shots to 0.005 at 10,000.

The exact probability is P(1) = 0.5. For each shot count, 40 independent Aer runs were made with distinct seeds, and the estimate $\hat{p} = K/N$ was recorded each time. The table compares the empirical RMSE of those 40 estimates with the predicted standard error.

Shots	Empirical RMSE	Theoretical SE	Mean absolute error
100	0.0518	0.0500	0.0408
500	0.0208	0.0224	0.0152
1000	0.0167	0.0158	0.0132
5000	0.0066	0.0071	0.0051
10000	0.0050	0.0050	0.0036



The empirical RMSE lies within 8% of the predicted standard error at every shot count. Between 100 and 10,000 shots the RMSE falls by a factor of 10.3 while the shot count rises 100-fold, which corresponds to a slope of about −0.51 on the log-log plot, close to the theoretical −0.5. Small departures such as the slightly lower value at 500 shots and higher value at 1000 are expected with only 40 repeats per shot count.

The mean estimates (0.494, 0.498, 0.499, 0.500, 0.500) approach 0.5 as shots increase, so the estimator shows no systematic bias. A single run need not show a smaller error than a run with fewer shots, which is why repeated experiments were used.

8. Consolidated verification summary

All deterministic verification checks passed with tolerances of 10^{-12} and far smaller actual deviations.

Metric	Value
Maximum gate unitarity residual	3.16×10^{-16}
Minimum fundamental-gate fidelity	0.9999999999999991

Metric	Value
Maximum fundamental probability L1 error	0
Minimum rotation fidelity	0.9999999999999991
Maximum rotation probability error	0
Minimum sequence theory/simulation fidelity	0.9999999999999991

The smallest fidelity, $1 - 9\times10^{-16}$, is a floating-point rounding effect: it equals about four units of double-precision machine epsilon (2.2×10^{-16}). Finite-shot results are not part of this table because they are random and are assessed statistically in Section 7.

9. Discussion and conclusion

The results answer the four research questions posed in Section 1.4.

1. **Do Qiskit statevectors reproduce analytically calculated gate transformations?** Yes. All 16 gate-and-input cases, 438 rotation points and 12 sequence runs matched NumPy with fidelity at least 0.9999999999999991 and probability error 0.
2. **How do Rx(θ), Ry(θ) and Rz(θ) change Bloch coordinates and measurement probabilities?** Each rotates the Bloch vector about its own axis. Rx and Ry change the probabilities of $|0\rangle$ as $\sin^2(\theta/2)$, while Rz leaves them unchanged but rotates $|+\rangle$ around the equator.
3. **Which gate sequences are order dependent, and can computational-basis measurements distinguish their outputs?** H with Z and X with H are order dependent, giving orthogonal states. Computational-basis measurements do not always distinguish them: on $|0\rangle$ both orders of each pair give P(1) = 0.5, although on $|+\rangle$ the probabilities differ by 1.
4. **How do finite-shot probability estimates converge toward ideal probabilities?** The RMSE tracks the binomial standard error, falling by a factor of 10.3 over a 100-fold increase in shots, close to the N^(−1/2) law.

9.1 Phase and population

X and Y change the computational-basis populations of $|0\rangle$ and $|1\rangle$, and H changes them for $|0\rangle$, $|1\rangle$ and $|+\rangle$. Z and Rz change only the relative phase. For $|0\rangle$, $Rz(\theta)|0\rangle = e^{(-i\theta/2)}|0\rangle$ differs from $|0\rangle$ by a global phase, so P(1) stays 0 even though the statevector changes. On

$|+\rangle$ the same phase becomes a relative phase and the state moves around the equator. Relative phase is therefore observable once the state is rotated into a basis where it affects the probabilities, as H does to $|+\rangle$ and $|-\rangle$.

9.2 Verification is not quantum advantage

The results confirm known identities on a classical simulator. A single qubit is cheap to simulate classically, so nothing here demonstrates a quantum speed-up or hardware performance.

9.3 Limitations

- The circuits use a single qubit, so there is no entanglement.
- Ideal statevectors omit hardware errors, and the shot simulation models sampling statistics, not device noise.
- The gate families and initial states are fixed, so the findings do not automatically extend to arbitrary quantum channels.
- The finite-shot study uses 40 repeats per shot count, so individual RMSE values carry sampling variation of their own.

Adding amplitude-damping or depolarising noise and comparing fidelity with the ideal predictions is a natural extension.

9.4 Conclusion

The project connected the linear algebra of single-qubit gates to their simulated behaviour in Qiskit. The largest unitarity residual was 3.16×10^{-16} and the lowest state fidelity was $1 - 9\times10^{-16}$, so analytical and simulated results agree to floating-point precision. The most notable phase-sensitive result is that Z and Rz are invisible on $|0\rangle$ but move $|+\rangle$ to $|-\rangle$ and around the equator, and the most notable order-dependence result is that H→Z and Z→H give orthogonal states with identical Z-basis probabilities on $|0\rangle$. Finite-shot errors followed binomial sampling theory. The framework extends naturally to multi-qubit gates, noisy simulation and variational circuits.

References

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